Project Workflow: Week 5

Phase: Road System & Player Car

1. Objective

The goal of Week 5 was to bring the player's car to life on a moving road. This included:

1. Implementing the 3-lane scrolling Road class with curve physics.
2. Implementing the abstract Car class and PlayerCar with lane-switching and speed boosting.
3. Wiring up Left/Right Arrow and Space key controls.

2. Detailed Task Breakdown

❖ Implementing the 3-Lane Scrolling Road

Built out the Road class to render a continuously scrolling 3-lane highway. The class tracks lane boundaries, scrolls road markings and edge textures downward each frame to simulate forward motion, and applies a simple curve-physics offset so the road can bend left or right over time.

❖ Implementing the Abstract Car Class and PlayerCar

Created an abstract Car class holding shared state and behavior, such as position, current lane, speed, and a base update()/draw() contract, so that PlayerCar and future EnemyCar subclasses stay consistent. PlayerCar extends Car and adds lane-switching logic that smoothly moves the car between the Road's three lanes, along with a speed-boosting mechanic that temporarily increases forward speed and scroll rate.

❖ Wiring Up Keyboard Controls

Added a KeyListener to GamePanel to capture Left Arrow, Right Arrow, and Space key events. Left/Right Arrow keys trigger PlayerCar's lane-switching method, and Space triggers the speed-boost method. Key states are tracked so the update() cycle from Week 4 can read player input every frame without blocking the Event Dispatch Thread.

3. Deliverables

1. **Road Class:** A 3-lane scrolling road with curve physics.
2. **Car Class:** An abstract base class defining shared car behavior.
3. **PlayerCar Class:** A working player car with lane-switching and speed boosting.
4. **Input Handling:** Left/Right Arrow and Space key controls wired into the game loop.

4. Tools & Resources

Development Environment

1. Java JDK 8+
2. IntelliJ IDEA / VS Code

- ### Reference Materials

Java KeyListener and key-binding documentation, Graphics2D transform references for road curve rendering, and object-oriented design references for abstract classes and inheritance.